Generative AI and Foundation Models

January 17, 2026

Navchetan Awasthi

Timings

- Every Saturday and Sunday
- 8:00 AM-09:30 AM
- Course slides will be uploaded after the lecture
- Lectures will also include written notes

Syllabus

- Will upload the old syllabus
- It will change according to what we can cover this year

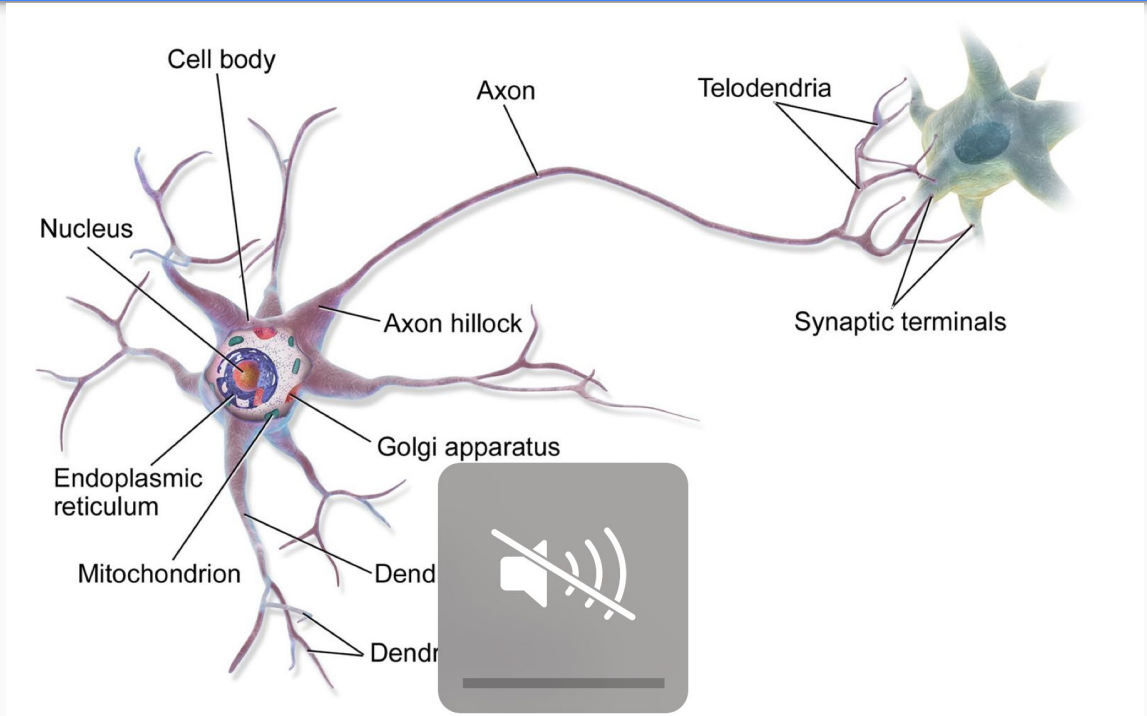
Marks Distribution

- Exams, Assignments, quizzes and project
- Assignments - 3 (Every month 1- from february 15th)
- Percentage I will let you know soon

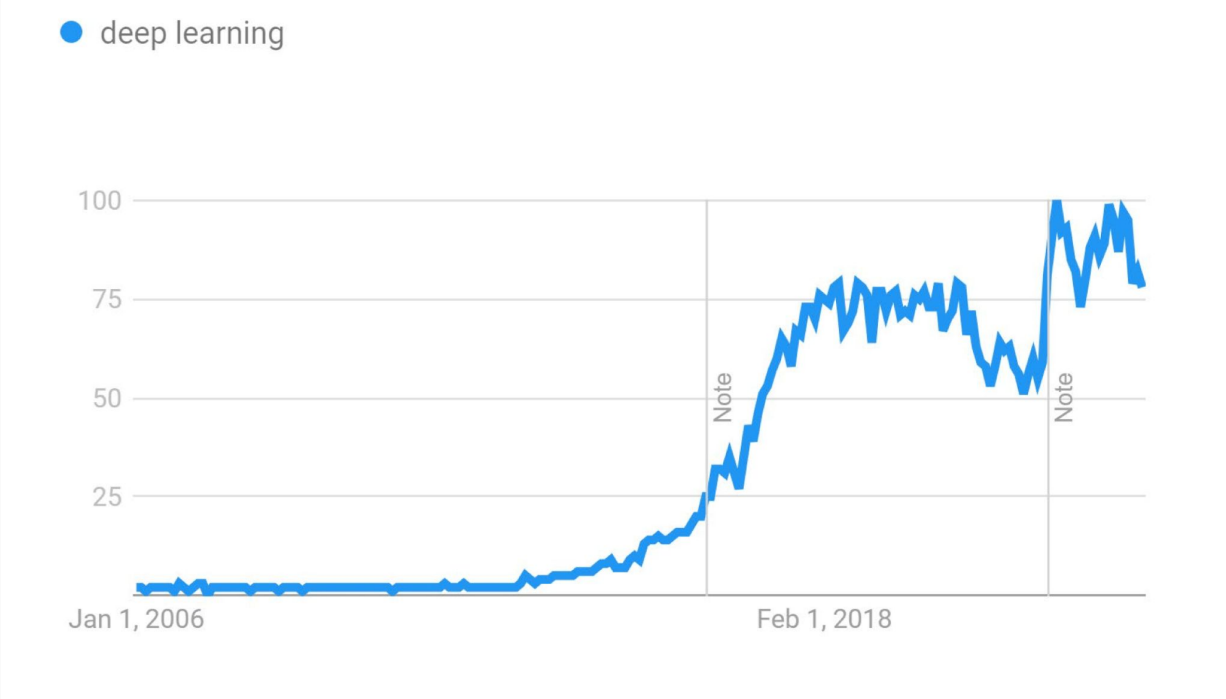
Deep Learning

- Deep learning attempts to learn representations of data with multiple levels of abstraction.
- Deep learning usually refers to a set of algorithms and computational models that are composed of multiple processing layers.
- These methods have significantly improved the state-of-the-art in many domains including, speech recognition, classification, pattern recognition, drug discovery, and genomics.

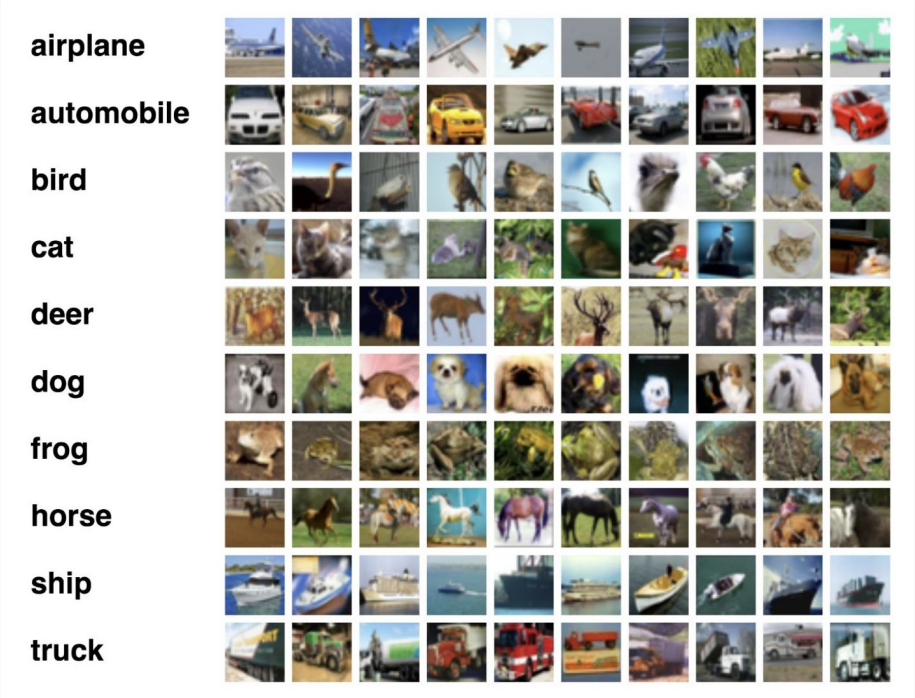
Biological Inspiration



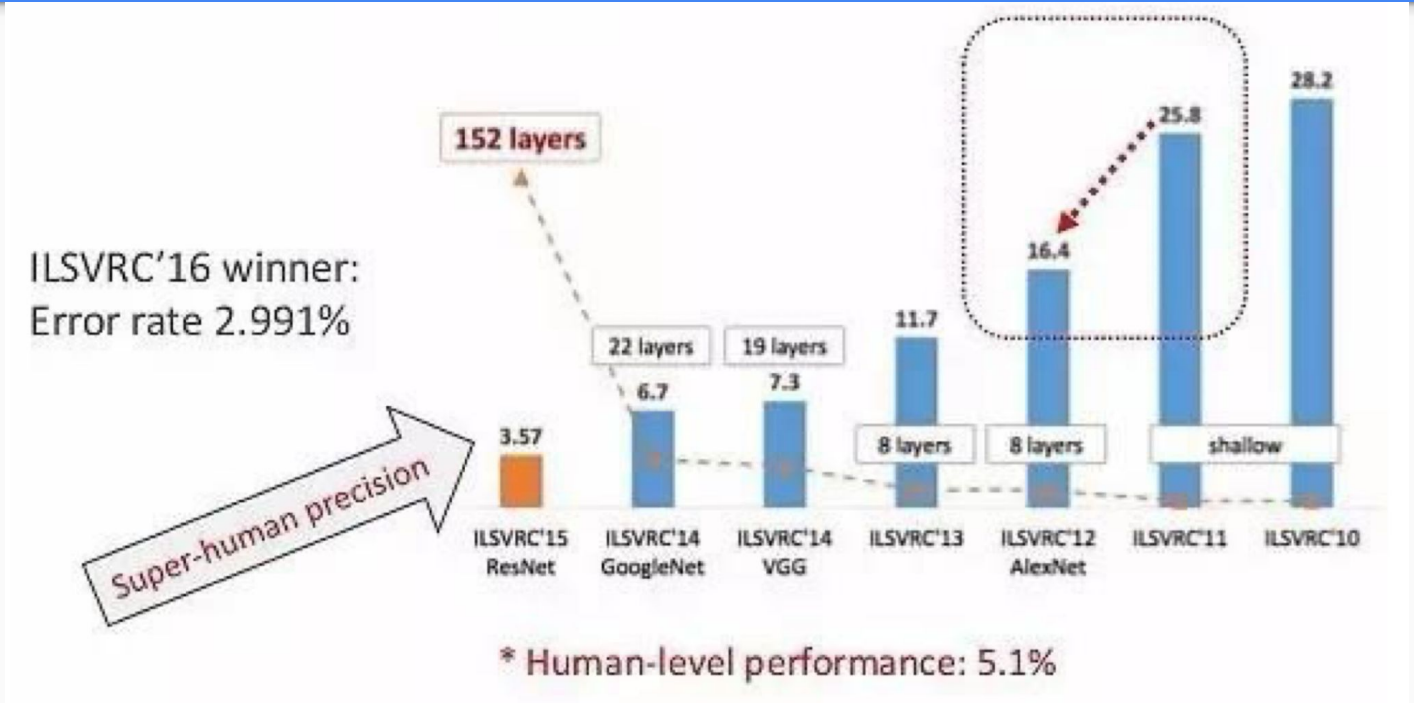
Interest over time



ImageNet

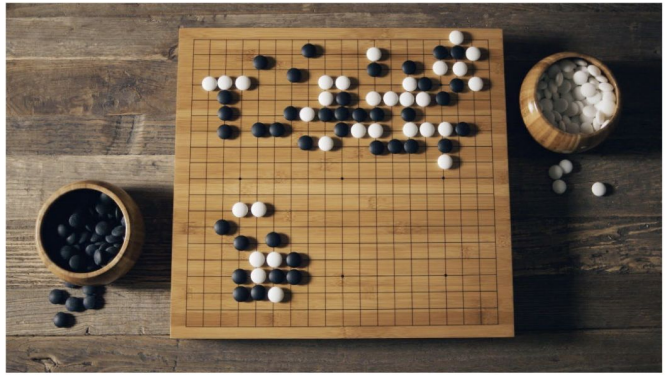

airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck

ImageNet challenge



AlphaGo

- AlphaGo (2016) algorithm masters ancient game of Go
- **Deep-learning software defeats human professional for first time.**



The Visual Microphone

https://www.youtube.com/watch?v=FKXOucXB4a8

Describing Photos

Generative Models

- Image Generation
- Data Augmentation
- Text Generation
- Music Generation
- Drug Discovery
- Speech Synthesis

Image Generation

https://thispersondoesnotexist.com/

ChatGPT

- Text Summarization
- Language Translation
- Code Generation
- Content Creation
- Customer Support
- Tutoring & Education
- General Q&A

Books written by AI

Alice and Sparkle Paperback – Jan. 14 2023

by Ammaar Reshi (Author), Chat GPT (Author), Mid Journey (Illustrator)

3.1 ★★★☆☆ 30 ratings

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Audiobook	Hardcover	**Paperback**
$0.00	$17.35 prime	**$11.95** prime
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The children's book covered in The Washington Post, TIME Magazine, Le Figaro, NBC News, and media worldwide.

This is a story about a young girl named Alice who discovers the magic of artificial intelligence. She creates her own AI, named Sparkle, and together they go on adventures and use their combined knowledge to make the world a better place. The story explores the incredible abilities of AI and the importance of using them for good. It is a tale of friendship and exploration, filled with magic and wonder.

Images from natural language descriptions

- Midjourney generate images from natural language descriptions.
- Similar to DALL-E and Stable Diffusion.

What we will cover in the course ?

Introduction to Foundation Models

- **Transformers & Attention:** Introduction to Transformers, Self-Attention, Cross-Attention, Perceiver: Arbitrary IO with transformers, Self Attention & Non-Parametric Transformers
- **Diffusion:** Sampling, Diffusion Models, Latent Diffusion Models (LDM)
- **Applications:** LLMs, LVMs and Applications in Vision

Foundation Model Architectures

- **Foundation Model Architectures:** Transformers in Language including GPT, BERT, RetNet, State Space Models
- **Transformers in Vision:** ViT, MLP-Mixer, Conformer, Vision-Language (VL) models, CLIP
- **Architectures:** Dual-Encoder, Fusion, Encoder-Decoder, Adapted LLM, Unified Architectures

What we will cover in the course ?

Multi-Modal & Generative Foundation Models

- **Training & Inference:** Training Objectives, Contrastive Learning, Efficient Inference Techniques, Pre-training, Fine-tuning & Parameter Efficient Fine-tuning (LoRA, QLoRA), Flash Attention, Retrieval Augmented Generation (RAG)

Evaluation & Benchmarking

- **Evaluation Protocols and Standard Benchmarks, Hallucinations, Bias & Fairness, Privacy, Memorization, Machine Unlearning, Prompt Engineering**

Books

Text and Reference Books

1. **Foundation Models for Natural Language Processing – Pre-trained Language Models Integrating Media**, Gerhard Paal, Sven Giesselbach, Springer 2023
2. Research papers will be shared in class

Thank you

Questions ?